

MAT1375, Classwork3, Fall2026

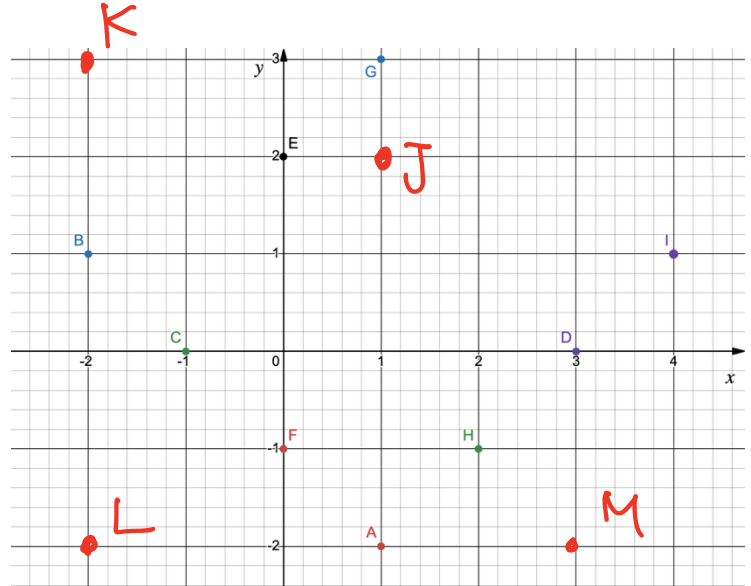
Ch3. Functions via Graphs

1. The definition of a Relation:

A relation is any set of ordered pairs. The set of all first components of the ordered pair is called the domain of the relation and the set of all second components is called the range of the relation.

2. Write down the ordered pair points in the given coordinate:

A: (1 , -2); B: (-2 , 1); C: (-1 , 0);
 D: (3 , 0); E: (0 , 2); F: (0 , -1);
 G: (1 , 3); H: (2 , -1); I: (4 , 1).



3. Plot the given points in a rectangular coordinate:

J:(1, 2); K:(-2, 3); L:(-2, -2); M:(3, -2).

4. From a relation to a function:

Each element in domain only gets one element from the range.

5. Linear Functions and the Slopes:

A linear function is a function of the form

$$f(x) = \underline{mx + b}$$

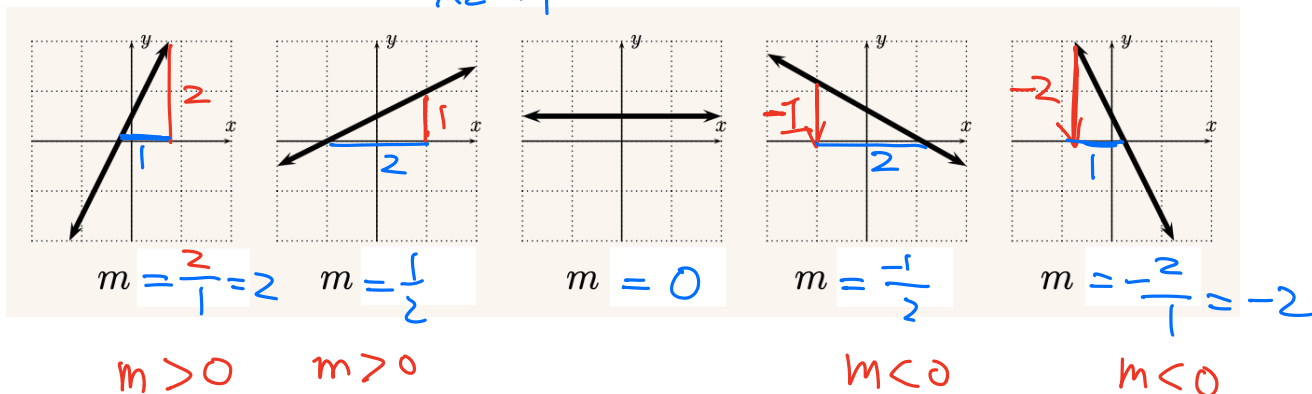
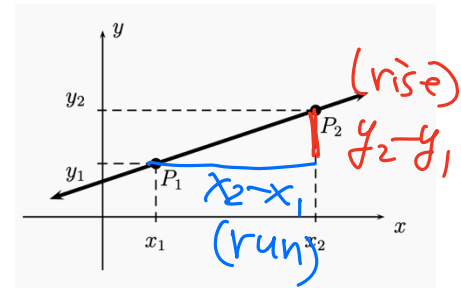
where m is the slope of the line and $(0, b)$ is y-intercept of the line. The domain of a line is All Real number, $\mathbb{R} = (-\infty, \infty)$

$$\mathbb{R} = \{x \mid x \in (-\infty, \infty)\}$$

6. The slope and its sign:

Given two points of a line $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. Then the slope m is

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



7. Find the equation of the line by the given information:

(1) $m = 2$ and y -intercept is $(0, -1)$

$$y = 2x + b$$

$$-1 = 2 \cdot 0 + b \Rightarrow b = -1$$

$$\Rightarrow y = 2x - 1$$

(2) the line passes $(1, 1)$ and $(-1, -3)$

$$m = \frac{3 - (-1)}{-1 - 1} = \frac{-4}{-2} = 2$$

$$y = 2x + b \Rightarrow y = 2x - 1$$

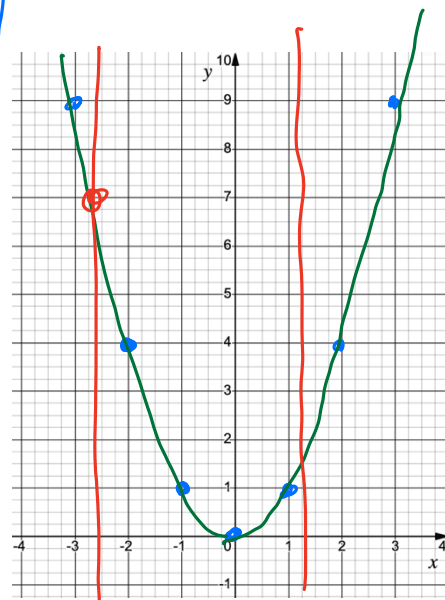
$$1 = 2 \cdot 1 + b$$

$$1 = 2 + b$$

$$-2 - 2$$

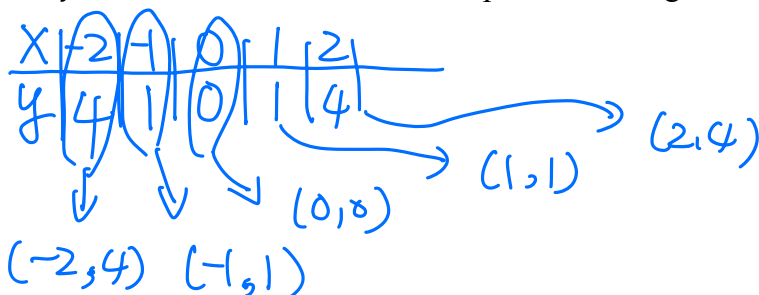
$$-1 = 0 + b$$

$$\Rightarrow b = -1$$



8. Functions given by graphs

Let $y = x^2$ with domain $D = \mathbb{R}$. Graph this on the given coordinate.

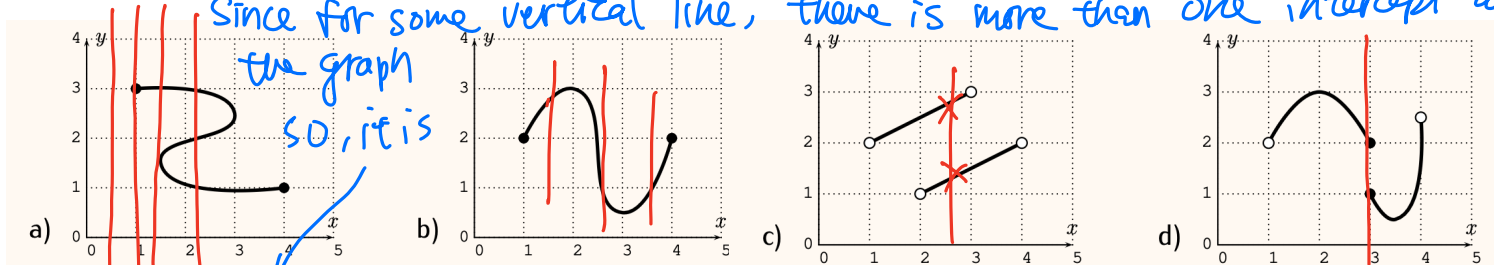


9. Vertical Line Test:

A graph is the graph of a function precisely when every vertical line intersects the graph once.

10. Use **Vertical Line Test** to determine which of the following are the graphs of functions.

Since for some vertical line, there is more than one intercept with the graph so, it is



NOT a function

Yes, it's a function

NO, not a function

NO, not a function

11. Let f be the function given by the following graph.

(a) What is the domain of f ?

$$(-3, 4) \cup (4, 7]$$

(b) What is the range of f ?

$$R_f = \{y \mid y \in (-2, 2]\}$$

(c) For which x is $f(x) < 0$?

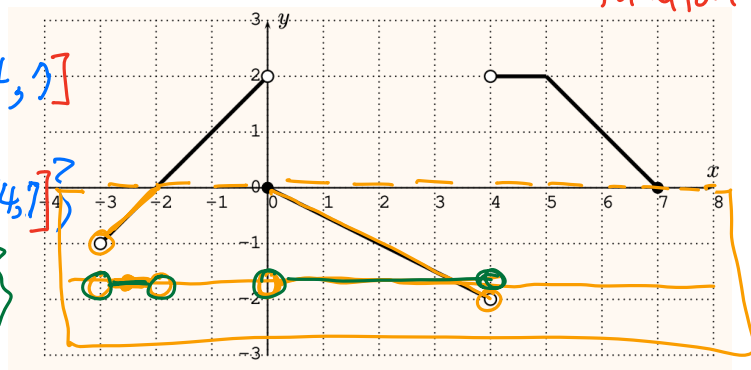
$$\{x \mid x \in (-3, -2) \cup (0, 4)\}$$

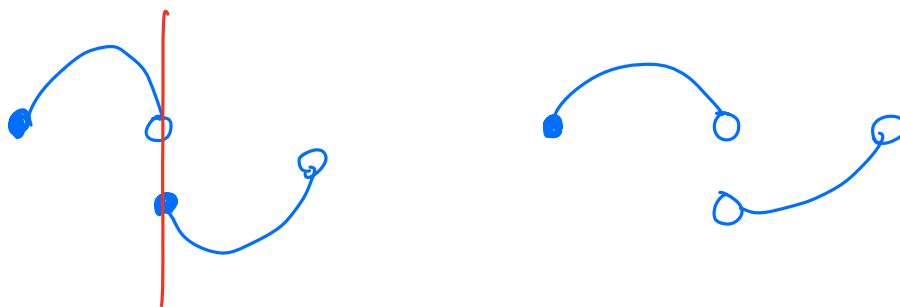
(d) Find $f(0) + 5$.

$$f(0) + 5 = 0 + 5 = 5$$

(e) Find $f(0 + 5)$.

$$f(0 + 5) = f(5) = 2$$





$[-3, 1]$